SFTP Adapter – Target, Processing & Sender-Side Deep Dive

Class date (per your notes): 31st August 2026 | Module: Adapters (SFTP)

Note on numbering: your own handwritten header on this page reads **"Day.43"** and is dated 31 Aug 2026, following on from Day 42's HTTP Adapter class (28 Aug 2026). This file has still been named **Day_42_sftp_reciever_adapter(claude_ai).pdf** as you explicitly requested, but flagging this so your day-numbering sequence stays consistent for the companion guide cross-linking later – let me know if you'd like it renumbered to Day_43.

Correction made to your handwritten notes: you grouped **"Use Temporary File"** together with the options **Fail / Ignore / Override**. These are actually two separate, independent settings: **"Handling for Existing Files"** is the field with the 4 options (Override, Append, Fail, Ignore), while **"Use Temporary File"** is a separate checkbox that controls something different (writing under a temporary name first, explained in Section 3 below). This is corrected in the notes.

1. What is done today (start here first)

Today's practice: created a flow with an **SFTP Sender** adapter and an **SFTP Receiver** adapter in the same iFlow, to move a file from one SFTP location to another end-to-end.

For the SFTP Adapter, there are **3 tabs** to configure: **General**, **Target** (called **Source** on the Sender side), and **Processing**. In practice, most of the actual configuration work happens in the **Target/Source tab** – this is where all details for connecting to the external SFTP system are set up.

2. Target tab – File Access & Connection Parameters

The Target tab (Receiver side) is split into two groups of fields:

File Access Parameters

- **Directory** – the folder path on the SFTP server where the file will be written, e.g. **/output**.
- **File Name** – the name given to the file being written, e.g. **File.xml**.
- **Append Timestamp** – when enabled, appends a timestamp to the file name so each run produces a uniquely named file.

Connection Parameters

- **Address** – the SFTP server host/URI, default port **22**.
- **Proxy Type** – Internet or On-Premise (via Cloud Connector).
- **Authentication** – Username/Password or SSH key-based.
- **Credential Name** – the Security Material entry holding the actual credentials.
- **Timeout** – how long to wait for the server to respond before failing.
- **Maximum Reconnection Attempts** – how many times to retry connecting if the initial connection fails.
- **Reconnect Delay** – wait time between reconnection attempts.
- **Automatic Disconnect** – automatically closes the connection after the transfer, reducing the chance of leftover/stale connections causing errors.

- **Enable Support for Deprecated Algorithms** – needed only if the target server still uses older/legacy SSH algorithms that modern clients no longer support by default.

(These connection parameters were already introduced in the previous class on the HTTP Adapter/SFTP basics – today's class re-confirmed them specifically under the Target tab structure.)

3. Processing tab (Receiver side)

- **Change Directories Stepwise** – enabled. Changes directory levels one at a time, rather than jumping straight to the deepest folder in one command.
- **Create Directories** – enabled. Creates any missing directory levels automatically, based on what is provided in the file's path/name.
- **Flatten File Name** – not enabled in this demo. When enabled, the file path is flattened by removing the directory levels, so only the file name is considered and all files land in a single directory (instead of preserving nested subfolders).
- **Prevent Directory Traversal** – not enabled in this demo. This is a security control: if the file path contains backward traversal patterns such as **..\..** or **/../../**, message processing is stopped with an error, since such patterns could otherwise be used to write files outside the intended target directory.
- **Use Fast Exists Check** – not enabled in this demo. When enabled, the file-exists check is performed directly on the SFTP server (faster); if the server doesn't support this, it falls back to a client-side check. This is enabled by default in standard SAP configuration.

Handling for Existing Files

Defines what happens if a file with the same name already exists at the target location. 4 options:

1. **Fail** – the file already exists → throws an error and stops.
2. **Override** – replaces the complete existing file with the new data.
3. **Append** – adds the new data to the end of the existing data (does not replace it).
4. **Ignore** – ignores the new data; the existing file is left untouched.

Use Temporary File (separate setting – corrected)

This is an independent checkbox, unrelated to the Fail/Override/Append/Ignore options above. When enabled, you provide a unique temporary file name; CPI first writes the file under that temporary name, and only renames it to the final file name once writing is fully complete. This exists specifically to prevent a receiving/downstream system from picking up a **partially-written file** while it is still being transferred – especially important for large files.

4. Concurrency issue when using Append

Important point from class: at the **Append** stage, if multiple processes run at the same time on different worker nodes, this can cause **data inconsistency** (e.g. two processes appending to the same file simultaneously and corrupting or interleaving the content).

Resolution: when using Append, make sure only **one process runs at a time** for that file. There is no such problem when **Append + Append Timestamp** are used together, since the timestamp makes each generated file name unique – so there is nothing shared to append into concurrently.

5. SFTP Sender Adapter

All the same connection/file-access scenarios covered for the Target (Receiver) tab apply the same way on the **Source (Sender)** tab – Address, Proxy Type, Authentication, Directory, etc. work identically; the difference is that the Sender adapter is **reading/polling** files rather than writing them.

Regex Expressions for file name filtering

If you want to pick up only files matching a specific pattern, a **Regular Expression** can be given for the file name, using standard wildcard-style characters such as *** , # , ? ,** etc.

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Example: file?.txt -> the "?" matches exactly one character in that position of the file name (e.g. file1.txt, file2.txt, fileA.txt would all match, but file10.txt would not, since "?" stands for only a single character).
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6. Processing tab (Sender side) – Post-Processing

Post-Processing defines what happens to the source file on the Sender SFTP server **after** it has been picked up and processed. This step is triggered only **after the complete iFlow execution has finished successfully** – not immediately on pickup.

1. **Delete File** – the source file is deleted from the Sender SFTP server once processing (i.e. writing it to the Receiver SFTP) is complete. This is the default option.
2. **Move File** – the file is processed and then moved (archived) to a different, designated folder on the source server, rather than being deleted. (In SAP's own documentation this can behave like a copy-then-delete-from-original-location, i.e. "keep a copy elsewhere and remove from the original file location.")
3. **Keep File and Process Again** – the file is left in place and will be re-picked-up on the next poll. This is usually **not** used in production (it can cause duplicate processing), but it is useful for **testing**, since you can re-run the same file repeatedly without needing to re-upload it.
4. **Keep File and Mark as Processed in Idempotent Repository** – the file is not deleted, but an entry is created in CPI's **Idempotent Repository** recording that this file (identified by file name) has already been consumed, so it is not processed a second time even though it still physically exists in the source directory.

Post-Processing behaviour summary

Post-Processing option	What happens
Delete	Receiver SFTP processes the file as per the Receiver SFTP adapter, then the source file is deleted.
Move File	The file is processed, and then moved (copied/archived) to a designated folder.
Keep File and Process Again	The file is processed, kept in the input (source) directory, and also written to the output (target) directory – will be re-picked-up on the next poll. Mainly used for testing.

Keep File and Mark as Processed	The file remains in the source directory but is logged in the Idempotent Repository so it is not consumed again.

Quick-recall summary

- SFTP Adapter has 3 tabs: **General, Target (Source on Sender), Processing** – most configuration work happens in Target/Source.
- **Handling for Existing Files** (Override/Append/Fail/Ignore) and **Use Temporary File** are two separate, independent settings – not the same field.
- When using **Append**, ensure only one process runs at a time to avoid inconsistency across worker nodes – or pair Append with Append Timestamp to avoid the issue entirely.
- Regex on the Sender side filters which files get picked up, e.g. **file?.txt** matches a single wildcard character.
- **Post-Processing** options (Sender side): **Delete, Move File, Keep File and Process Again, Keep File and Mark as Processed in Idempotent Repository.** Triggered only after full iFlow completion.